

Data Warehouse and Data Lake

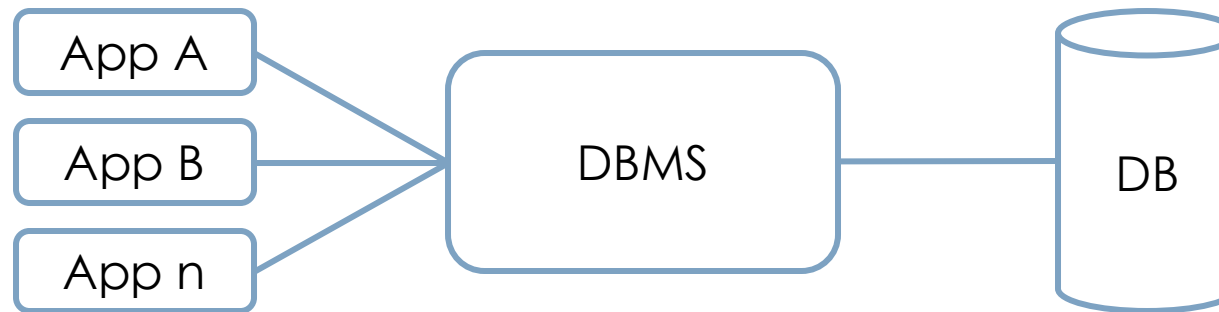
Irene Siragusa, PhD

Outline

-  Introduction
-  Resource and process
-  OLTP and OLAP
-  Data Warehouse
-  Data Lake

Introduction

- A database is a persistent and shared collection of data managed effectively, efficiently and trustworthily by a BDMS
- A company has a central and unique database in which data are organized and storing accordingly and are accessed by the target applications



Does it happen? No.

Introduction

- It is realistic? Of course, not
 - Continuous changing in applications and needs
- Generally, a company has DBs which are
 - *Multiple*
 - *Heterogeneous*
 - *Autonomous*

Resource and process

- Resource

- Everything that is managed by a company for its purposes
- Information and data are resources

- Process

- Sequences of decisions and actions taken by a company to pursue its objectives
- Processes involves managing the life cycle of resources

Process

- Operational process
 - Structural operations
 - Perfect defined rules
 - Sectorial and specific data
- Tactical process
 - Semi-structural operations
 - Known rules and human responsibility
 - Sectorial and partially aggregated data
- Strategic process
 - No rules
 - Highly aggregated data

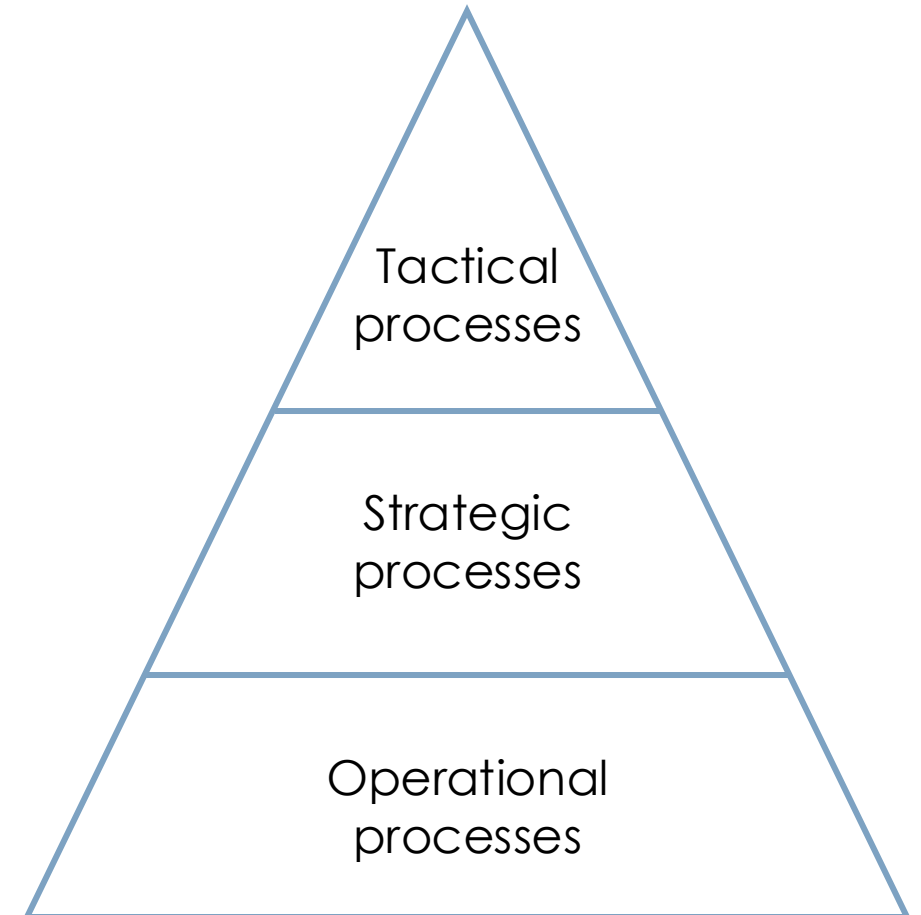
Transaction
processing systems

Management
information systems

Decision
support systems

Process

- Processes has a hierarchical structure
 - higher levels require a broader overview with aggregated data (tactical and strategic processes)
 - lower-levels require a more detailed focus or single record (operational process)



OLTP vs OLAP

Two different types of processing systems

- On-Line Transaction Processing (OLTP)
 - Traditional transactional operations
 - Pre-defined, fast and easy
 - Frequently made over small quantity of data
 - ACID properties
- On-Line Analytical Processing (OLAP)
 - Random and complex operations
 - Large quantity of data are involved
 - May require aggregated and historical data
 - On-read only operations (no ACID)

OLTP vs OLAP

	OLTP	OLAP
<i>User</i>	employee	manager
<i>Functions</i>	daily operations	support decision
<i>Project</i>	application-oriented	data-oriented
<i>Data</i>	current, detailed, homogeneous	historical, aggregated, heterogeneous
<i>Usage</i>	repetitive, simple queries	random, complex queries
<i>Access</i>	read-write	read
<i>Metric</i>	throughput	answer-time

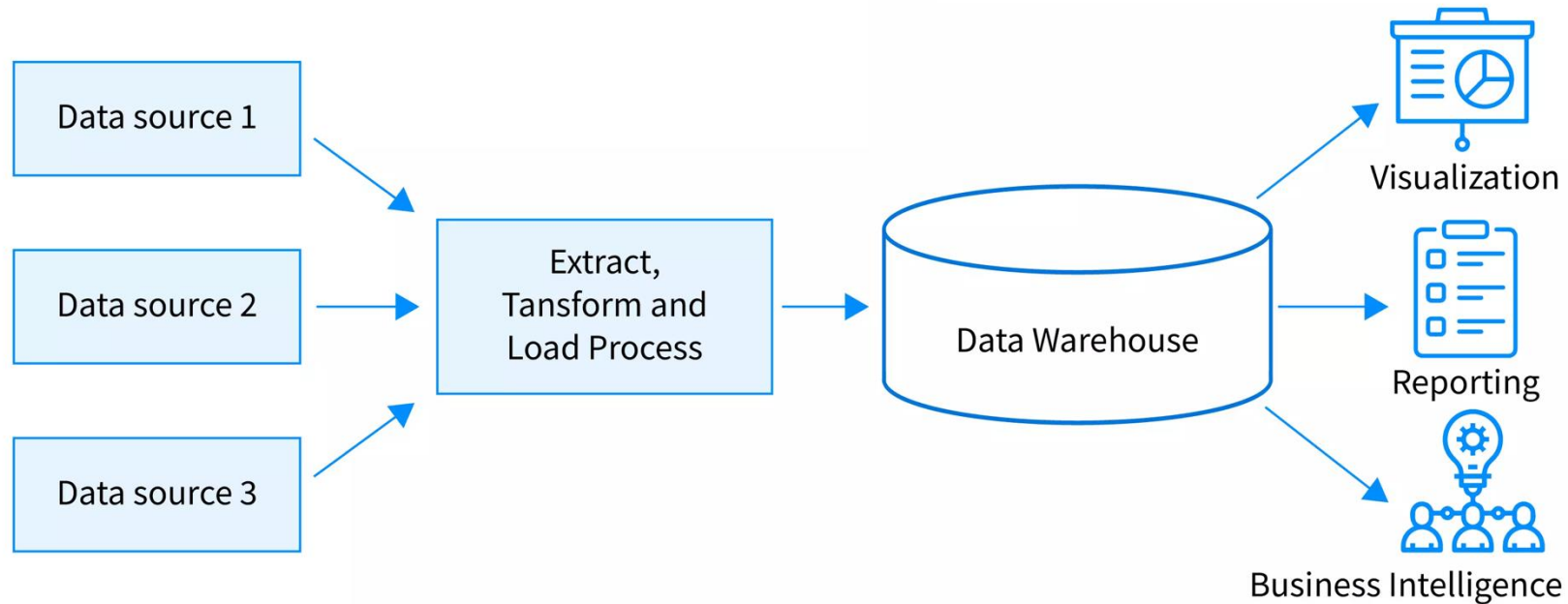
Data Warehouse (DW)

A Data Warehouse (DW) is a database

- mainly used for management decision support (*OLAP*)
- *Integrated* since involves the entire company (not single department)
- Structured in a *data-driven* way
- *Historical and aggregated data* are included
- Data are updated periodically and it is separate from operational databases

Data Warehouse

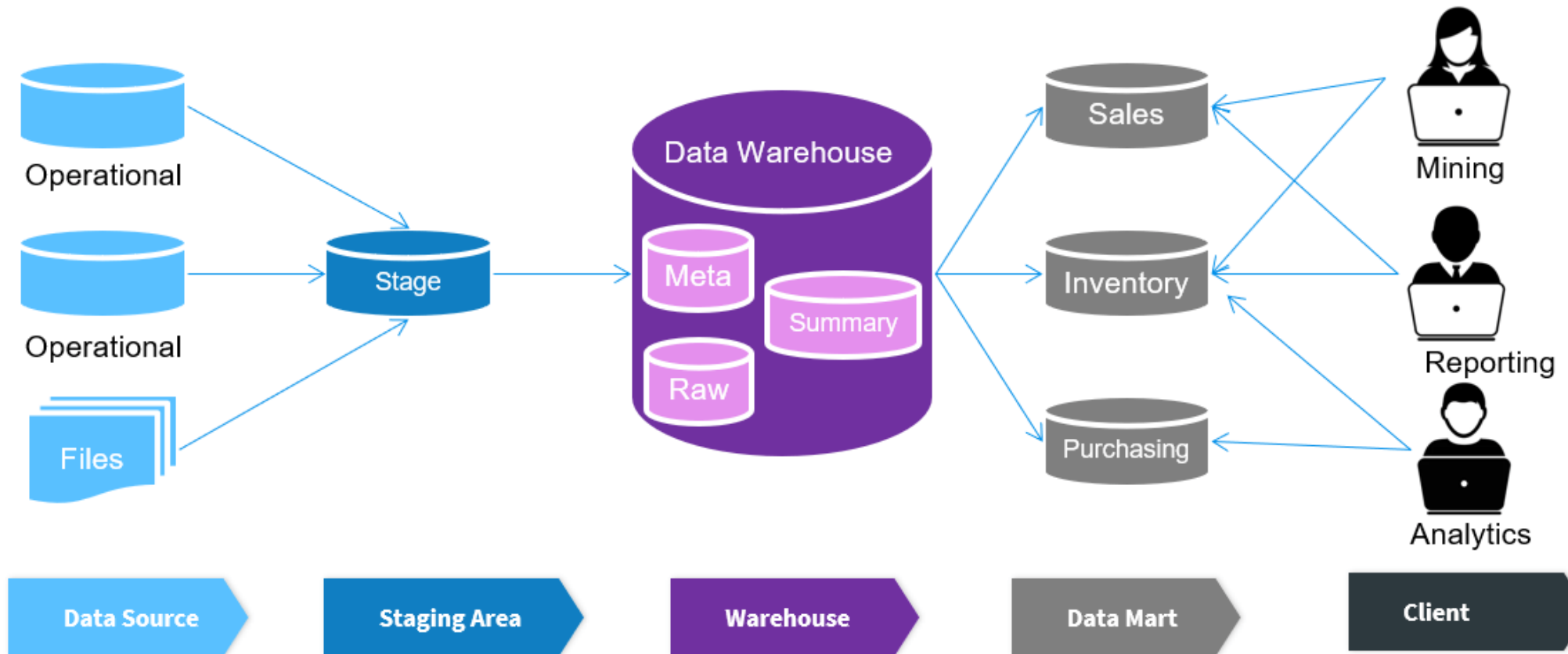
Data Warehousing



Data Warehouse

- A realistic architecture does not involve an integrated DW
 - Development costs
 - Not a priority
- Since a DW may be queried for specific processes and applications, a good tradeoff can be building a subset
- Data mart (DM)
 - It is a logical subset of a DW
 - Can be seen as a DW restriction to a single process
 - A DW can be seen (but not always) as a union of its data mart

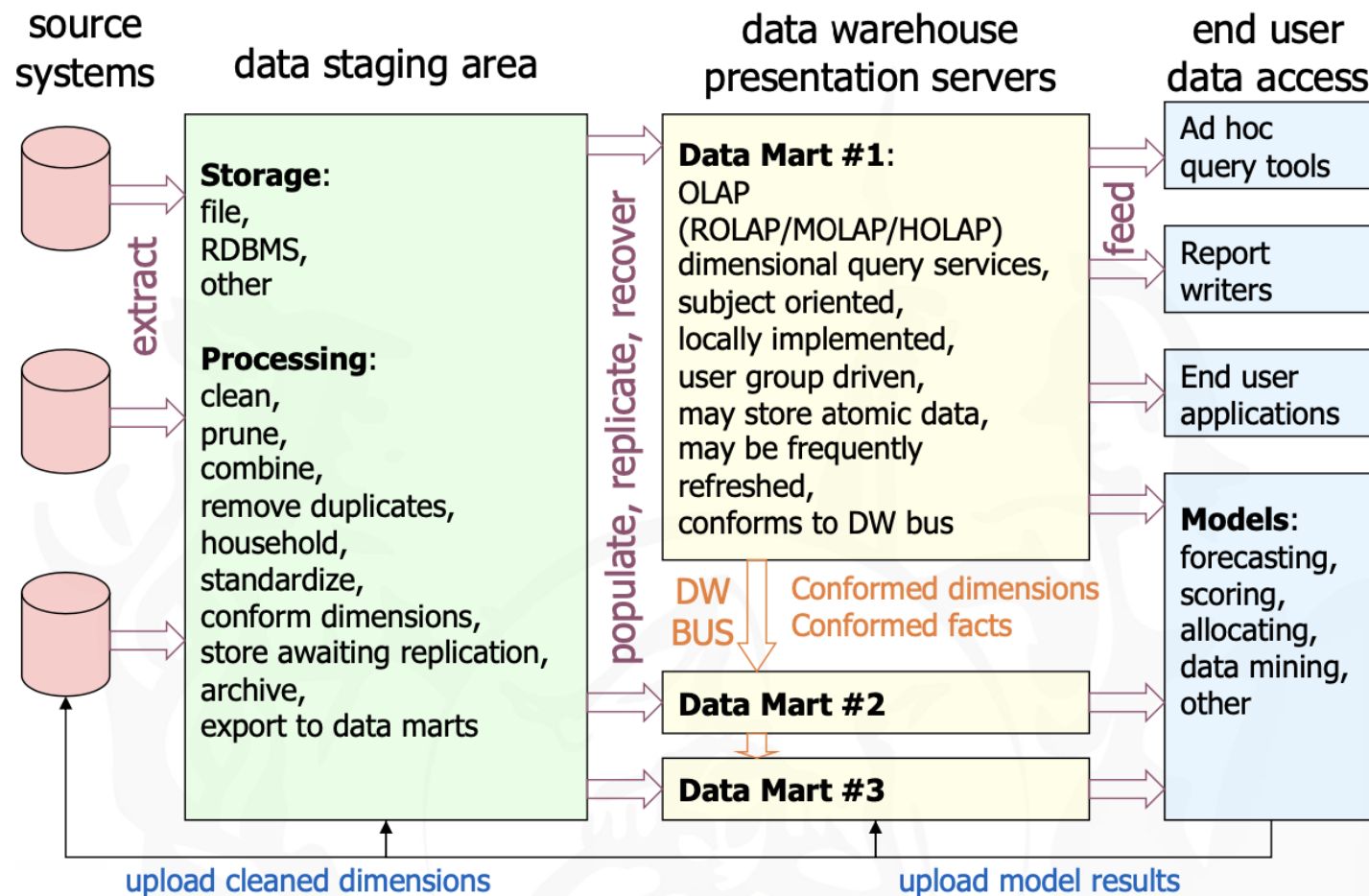
Data Warehouse



Data Warehouse vs Data Mart

- Top-down or bottom-up approach? Build DW or DM first?
- A DM is easier to built but sum of DMs do not make a good DW and building a DW from scratch is impractical.
- Usually, an incremental approach is used, and DMs coordination is necessary

Data Warehouse



Data Warehouse

Source Systems

- OLTP systems of the company that do not keep historical data and manages different processes and manages data differently
- External data

Data Warehouse

Data staging area

- Serves for data transit from source systems to DW
- Includes area for data extraction and pre-processing of the data, thus assuring data homogeneity, since input data are heterogenous
- Requires a complex set of simple tasks
- Is distributed over multiple nodes
- Performs ETL operations
 - Extract, Transform, Load

Data Warehouse

Presentation server

- system in which DW data is organized and stored for querying directly by end users, report writers, and other applications
- Data are represented in a multidimensional form (according to the concepts of fact and dimension)
- Technologies that can be adopted
 - Relational OLAP (ROLAP) and Multidimensional OLAP (MOLAP)

Data visualization

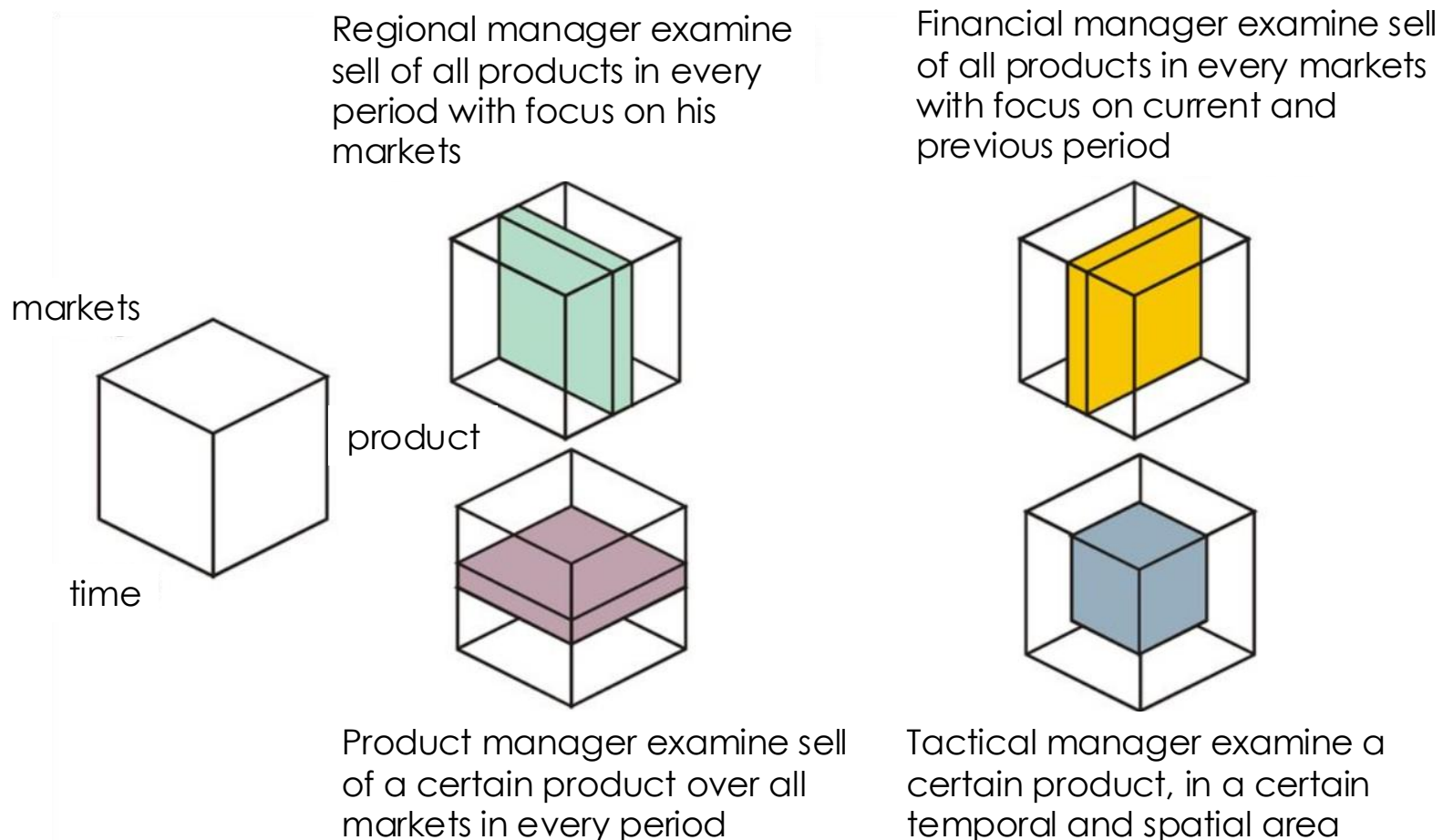
- Data are represented in a multi-dimensional form

Data Warehouse

Logical model

- Used to represent data in a graphical manner, assuring easier interpretability
 - Tables, histograms, plots, ...
- A fact is a focus concept of the analysis
- A measure is an atomic property of a fact
- A dimension is a perspective over which the analysis is performed
 - eg sales, qtn of sales and collection, product and time

Data Warehouse

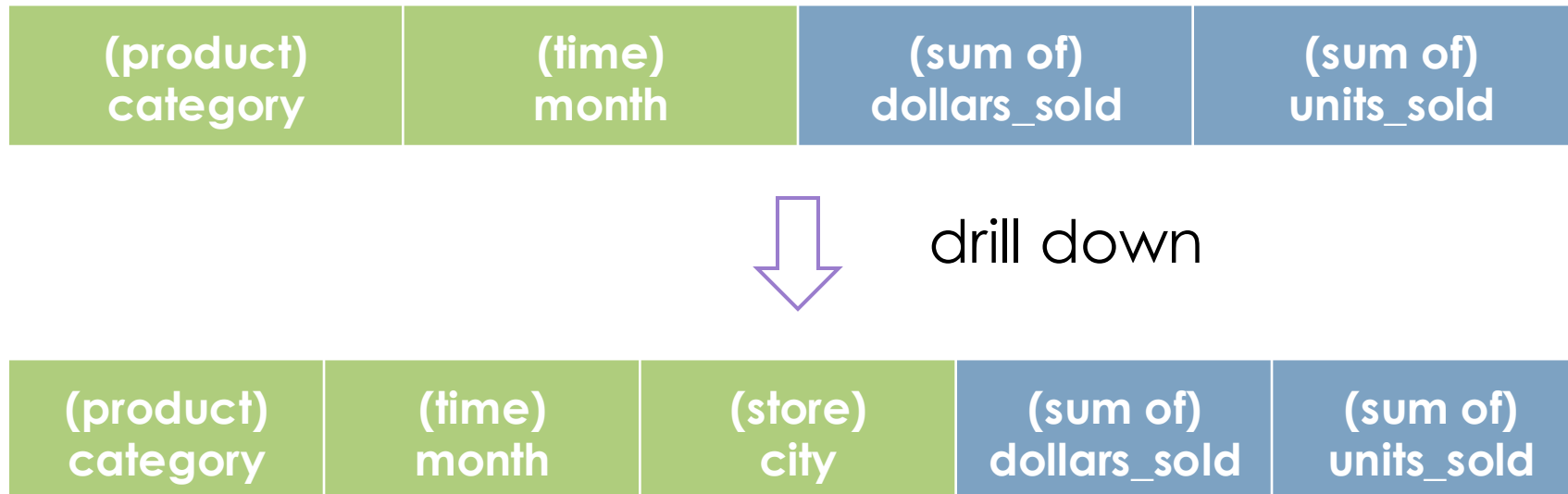


Data Warehouse

- Different operations can be done
 - Roll-up (drill up) to aggregate data
 - Drill down to dis-aggregate data
 - Slice & dice to select and project
 - Pivot
- Note that is necessary to determine if a fact (its measure) is additive, semi-additive or not additive in terms of aggregation

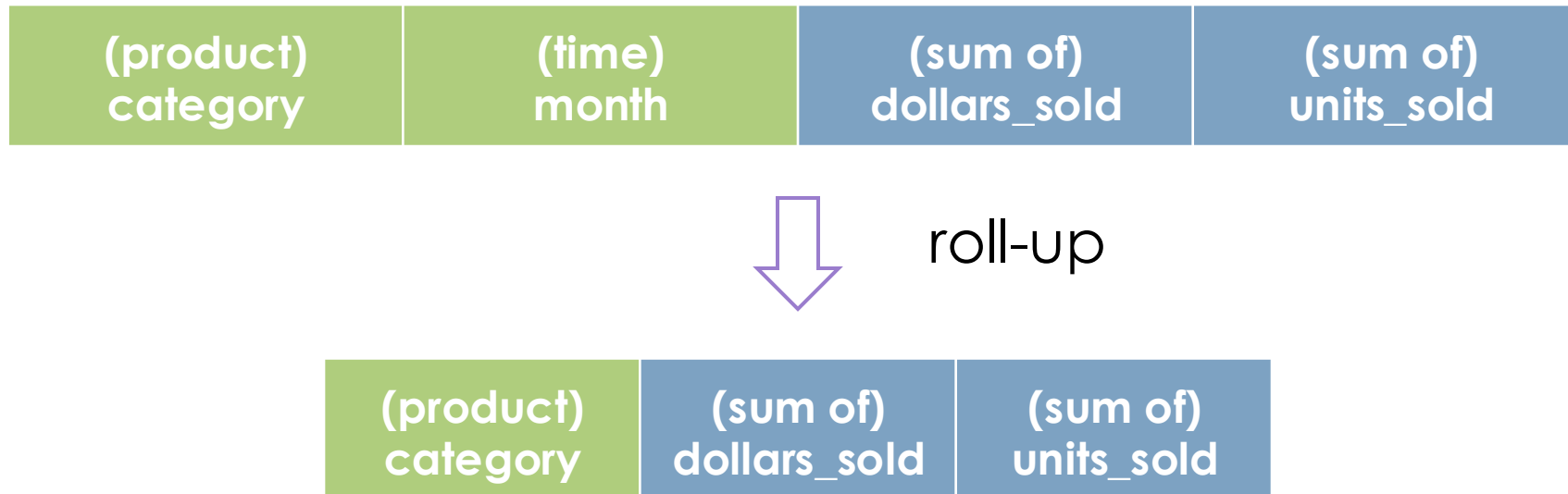
Data Warehouse

- Drill-down operation add details over returned data, which are fine grained



Data Warehouse

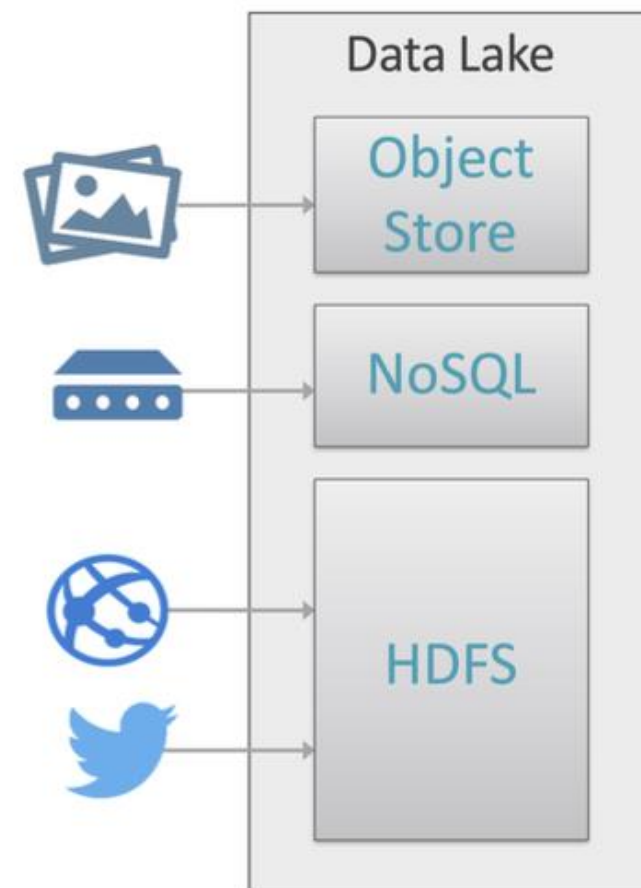
- Roll-up operation remove details over returned data, which are coarse grained



Data Lake

Data Lake (DL)

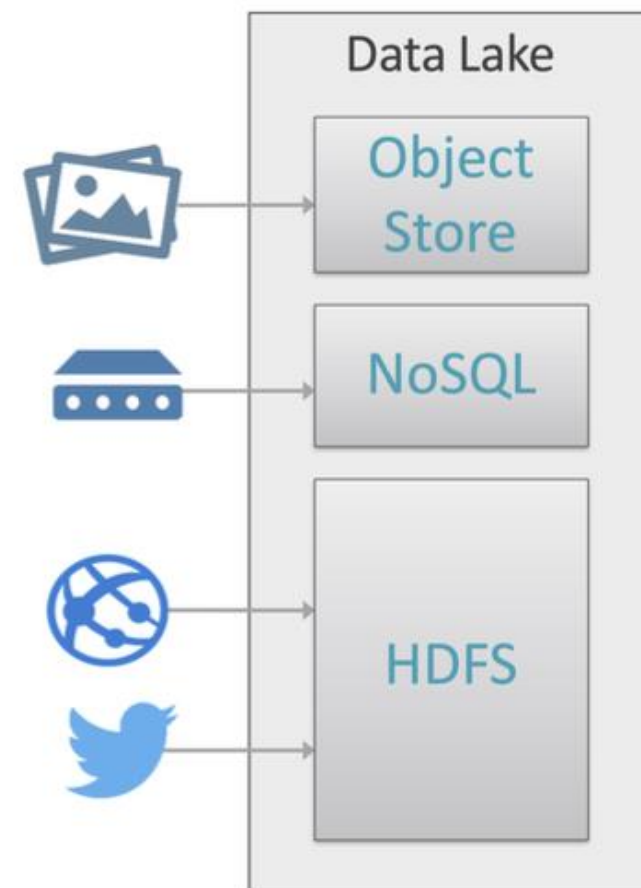
- It is a repository for heterogeneous data stored in their native format
- Speed up decision support systems
- Can be implemented via Big Data technologies
 - NoSQL, HDFS (Hadoop Distributed File System), object store for multidimensional data



Data Lake

Data Lake (DL)

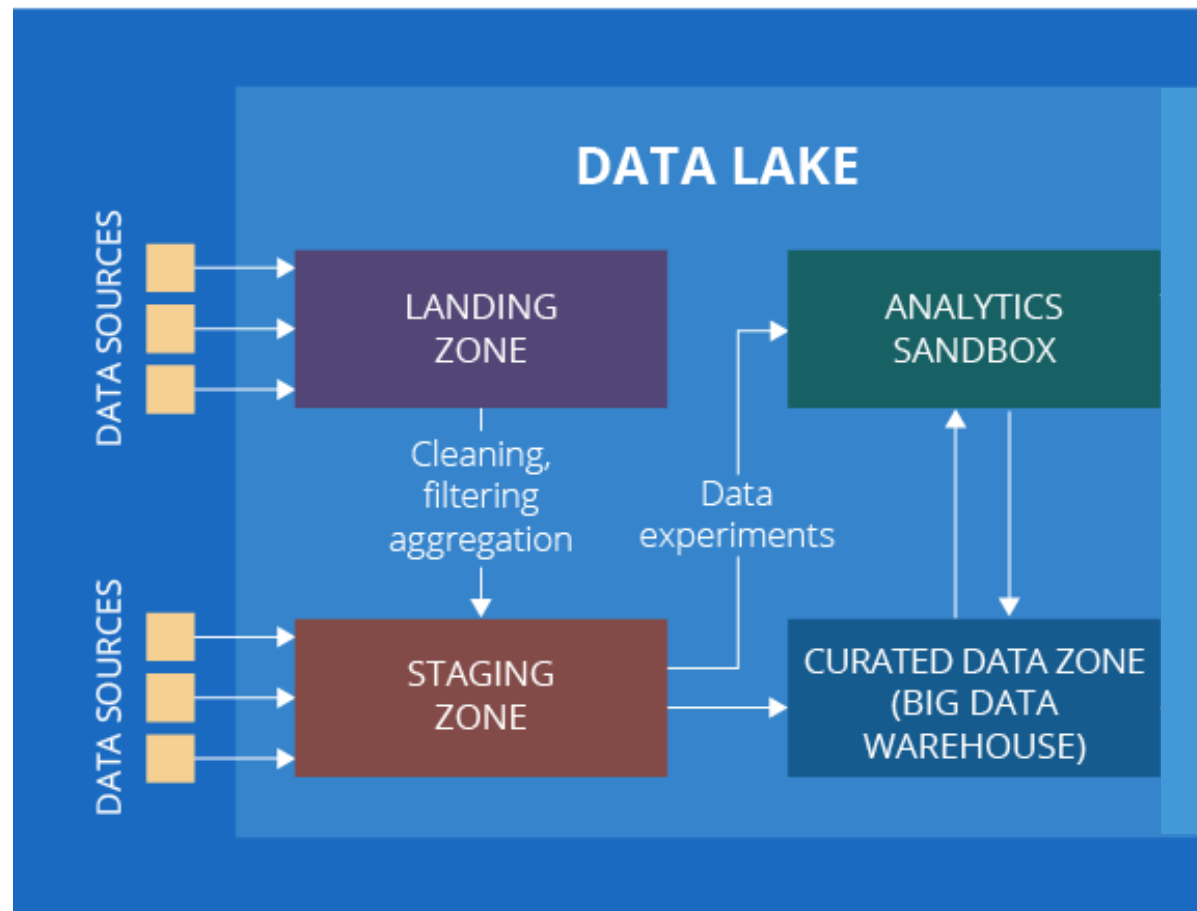
- No need for data schema
- New data can be acquired easily
- Able to manage large volume of data
- Support for data analytics functionalities



Data Lake

Landing zone

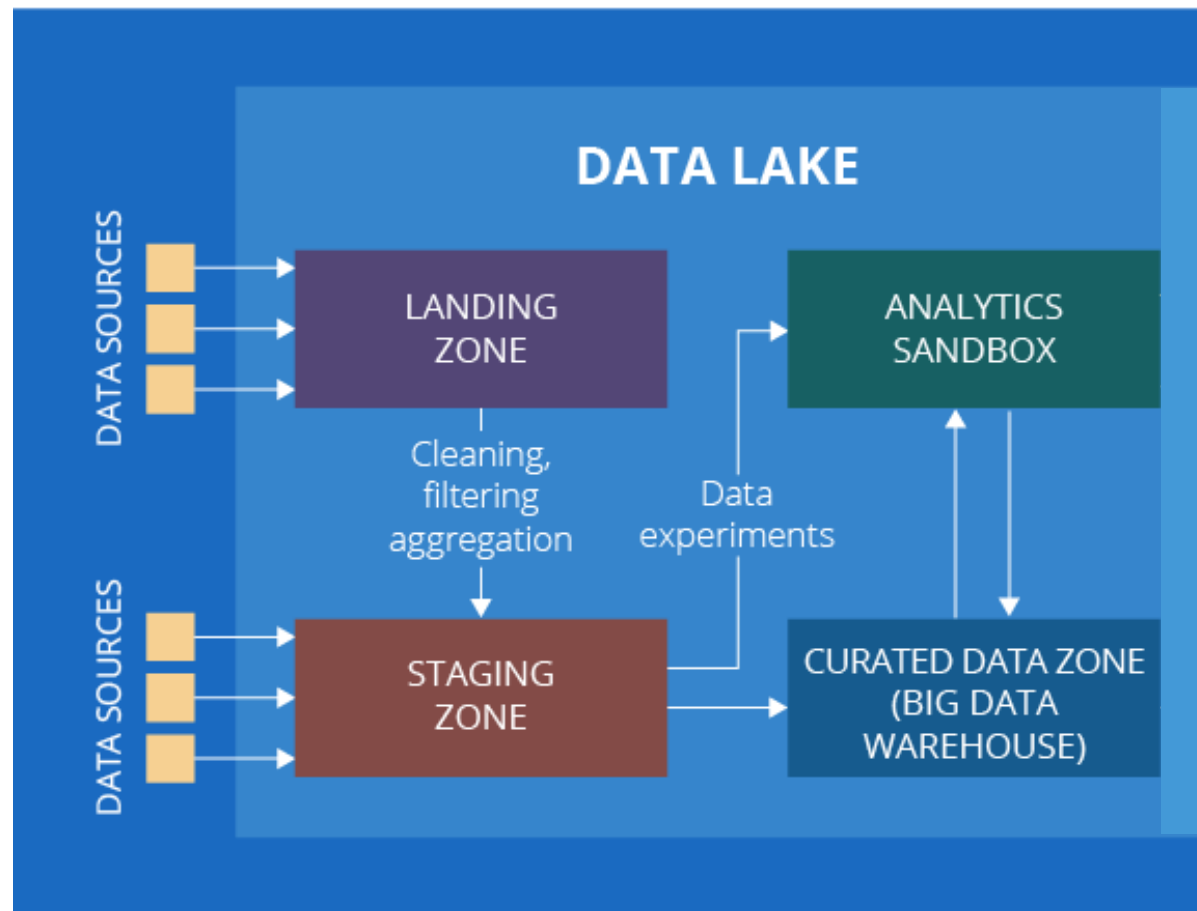
- Used if consistency and/or validity checks are required
- Checks are performed before data enter in the staging zone



Data Lake

Staging zone

- Every data type is supported
- Data history is preserved
- Data here not modified



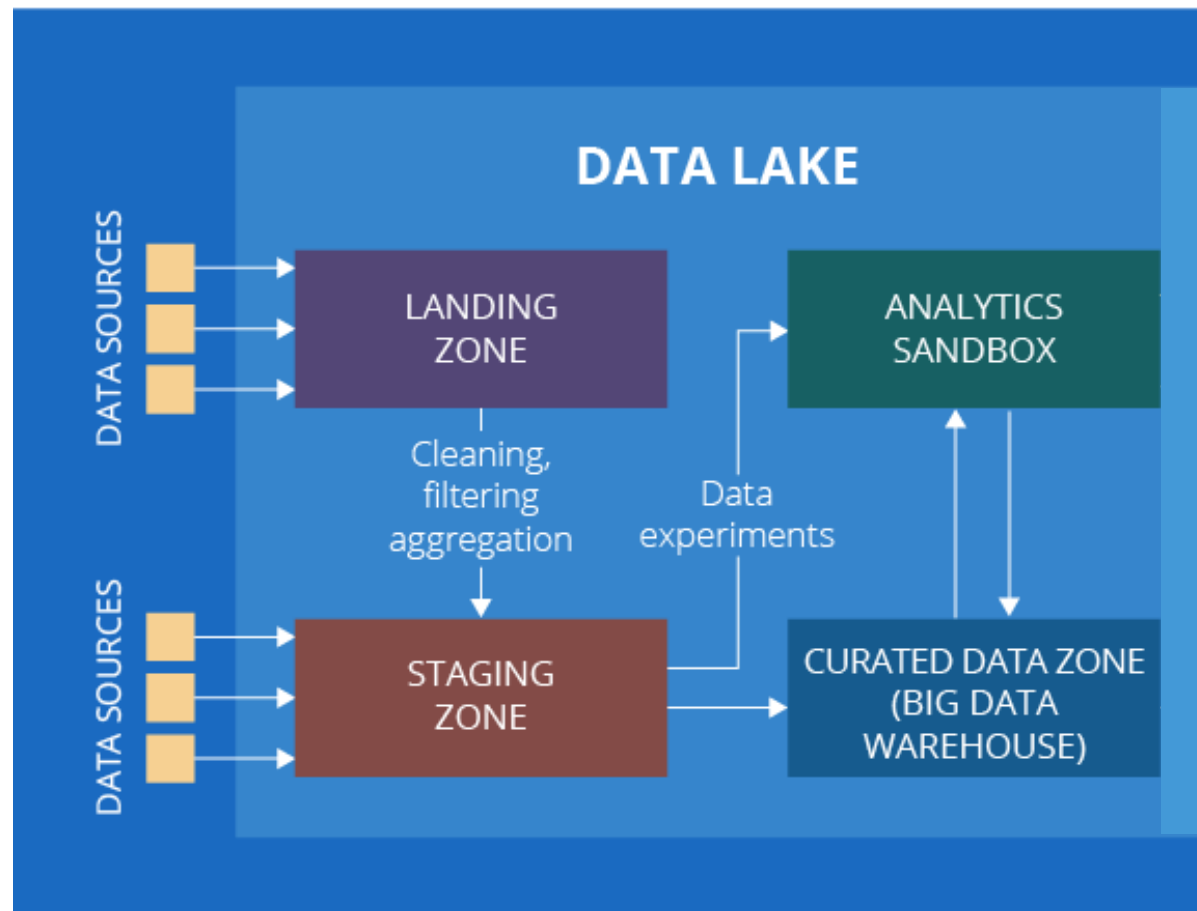
Data Lake

Curated Data zone

- Clean data to enable subsequent operations

Analytics Sandbox

- Area for exploration and data analysis process execution



Data Warehouse and Data Lake

- DL as staging area for DW
- DL as archive following DW politics
- DL as active component in analytics process

